

Answer Keys to Homework Problems

HW1:

Problem 1: a) -0.10384 C b) -0.19865 C c) -3.941 C d) -26.08 C

Problem 2: 25 C Problem 3: $q = 3,672 \text{ C}$ $E = 4406.4 \text{ J}$

Problem 4: a) 1.4725 C b) $-90e^{-4t}$ c) -22.5 J

Problem 5: 3A Problem 6: \$1.35 Problem 7: 24 cents

HW2:

Problem 1: a) 6 branches and 5 nodes. b) 7 branches and 5 nodes.

Problem 2: $i_1 = 14 \text{ A}$ $i_2 = -2 \text{ A}$ $i_3 = 10 \text{ A}$

Problem 3: $v_1 = 6$ $v_2 = 3 \text{ V}$ Problem 4: $v_1 = 2 \text{ V}$ $v_2 = -22 \text{ V}$ $v_3 = 10 \text{ V}$

Problem 5: $v_x = 4.167 \text{ V}$ Problem 6: $v_o = 8 \text{ V}$ and $I_o = 0.2 \text{ A}$

Problem 7: $R_{eq} = 727.3$ $R_{eq} = 3 \text{ K}$ Problem 8: R_1 and R_3

Problem 9: $R_1 = 3000$ $R_2 = \infty$

HW3:

Problem 1: $v_1 = 0 \text{ V}$, $v_2 = 24 \text{ V}$ Problem 2: $v_o = 20 \text{ V}$ Problem 3: $i_o = 1.73 \text{ A}$

Problem 4: $V_1 = 10 \text{ V}$ $V_2 = 4.933 \text{ V}$ $V_3 = 12.267 \text{ V}$

Problem 5: $R_L = 9$ Problem 6: $I_1 = 0.8$ $I_2 = -0.9$

Problem 7: $I_o = -1.7333 \text{ A}$ Problem 8: $i_o/i_s = -0.3$

Problem 9:
$$\begin{bmatrix} 9 & -3 & -4 & 0 \\ -3 & 8 & 0 & 0 \\ -4 & 0 & 6 & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \\ 2 \\ -3 \end{bmatrix}$$

Problem 10: $v_o/v_s = -8$

HW4:

Problem 1: $v_o = 4.5 \text{ V}$ Problem 2: $i = 1.875 \text{ A}$ $p = 10.55 \text{ watts}$

Problem 3: $v_x = -26.67 \text{ V}$ Problem 4: $I = 3 \text{ A}$ $P = 72 \text{ W}$ Problem 5: $v_o = 3 \text{ V}$

Problem 6: $v_o = -125 \text{ mV}$ Problem 7: $R_{th} = 20$ $V_{th} = -16.4 \text{ V}$

Problem 8: $R_n = 3$ $I_n = 1 \text{ A}$ Problem 9: $R = 25$ $P = 7.84 \text{ W}$

Problem 10: a) $R_s = 8 \text{ k ohms}$, $I_s = 10 \text{ mA}$ b) $i_m = 9.926 \text{ mA}$

HW5:

Problem 1: a) $v_o = 2V$ b) $v_o = 3V$

Problem 2: $v_o = 2.7V$ $i_o = 288 \mu A$

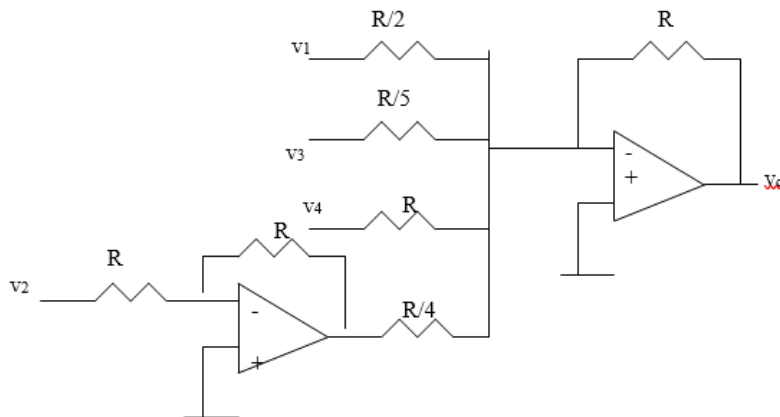
Problem 3: $i_o = 0.375 \text{ mA}$

Problem 4: $v_o = 1.8$

Problem 5: $P = 108 \text{ mW}$ $i_x = -6 \text{ mA}$

Problem 6: $R_f = 10k$ $R_1=R_2=R_3=R_4 = 40k$

Problem 7: $v_o = 14.09V$



Problem 8:

Problem 9: $i_o = 1.3695 \text{ mA}$

Problem 10: $i_o/i_s = 9$

HW6:

Problem 1: $v(2) = 13.624$ $p(2) = 70.66W$

Problem 2: $v_1 = 30V$ $v_2=40V$

Problem 3: a) $v_1(t) = -1250e^{-2t} + 1300V$ $v_2(t) = -250e^{-2t} + 270V$

b) $w_{12\mu F} = 4.235J$

$w_{20\mu F} = 0.3169J$

$w_{40\mu F} = 0.6339J$

Problem 4: a) $Leq = 6.667 \text{ mH}$ $i_1(t) = e^{-t} \text{ mA}$ $i_2 = 2e^{-t} \text{ mA}$

b) $v_o = -20e^{-t} \text{ uV}$ c) $w = 1.353 \text{ nJ}$

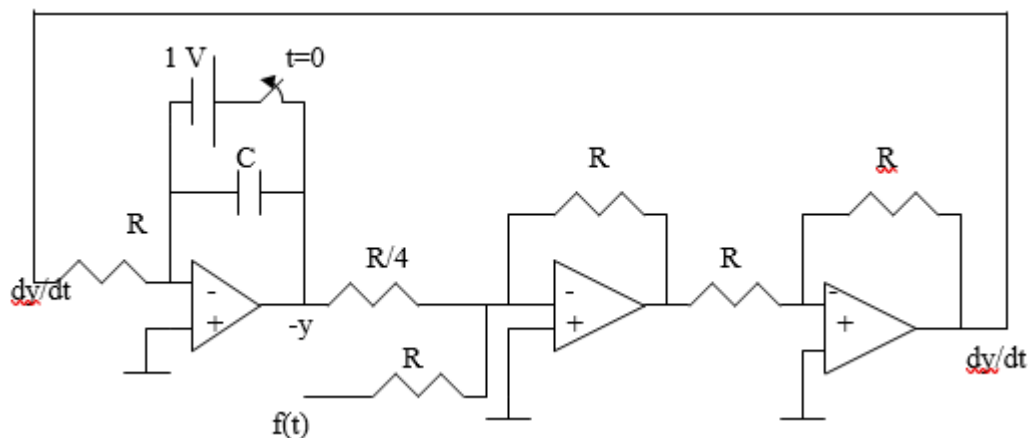
Problem 5: a) $w_5 = 40 \text{ J}$ $w_{20} = 40 \text{ J}$ b) $W = 80J$

c) $i_1 = 5 \times 10^{-5}(e^{-200t} - 1) + 4 \text{ A}$ $i_2 = 1.25 \times 10^{-5}(e^{-200t} - 1) - 2 \text{ A}$

d) $i = 6.25 \times 10^{-5}(e^{-200t} - 1) + 2 \text{ A}$

Problem 6: $v_o = -30 \text{ mV}$

Problem 7



HW7:

Problem 1: a) $R=7\text{ K}$, $C = 0.7143\text{ uF}$ b) $\tau_{aw} = 5\text{ ms}$ c) $\tau_o = 3.466\text{ ms}$

Problem 2: $v = 40 e^{-50t}\text{ V}$

Problem 3: $v_o(t) = 4e^{-5t}\text{ V}$ for all $t > 0$

Problem 4: $i_o(t) = 1.41e^{-3t}\text{ A}$

Problem 5: $i(t) = 2e^{-5t} u(t)\text{ A}$

Problem 6: a) $v(t) = 4$ for $t < 0$ and $v(t) = 20 - 16e^{-t/8}$ for $t > 0$ b) $v(t) = 4\text{ V}$ for $t < 0$ and $12 - 8e^{-t/6}$ for $t > 0$

Problem 7: $i(t) = 0.8\text{ A}$ for $t < 0$ and $i(t) = 0.8e^{-t/480}\text{ A}$ for $t > 0$

Problem 8: $v(t) = 96\text{ V}$ for $t < 0$ and $v(t) = 96e^{-4t} u(t)\text{ V}$

Problem 9: $i_o(t) = -50 u(t)\text{ mA}$

Problem 10: Speed = 6.278 m/s

HW9:

Problem1: (a) $V_m = \underline{50\text{ V}}$. (b) Period = 209.4 ms (c) Frequency $f = 4.775\text{ Hz}$. (d) $v(0.01) = 44.48\text{ V}$

Problem2: the phase angle between the two signals is $\underline{20^\circ}$ and $\underline{v_1 \text{ lags } v_2}$.

Problem3: a) $50.88 \angle -15.52^\circ$. b) $60.02 \angle -110.96^\circ$.

Problem4: (a) $V = 21 \angle -15^\circ\text{ V}$ (b) $I = 8 \angle 160^\circ\text{ mA}$ (c) $V = 120 \angle -140^\circ\text{ V}$ (d) $I = 60 \angle 190^\circ\text{ mA}$

Problem5: (a) $-1.2749 + j0.152$ (b) -2.083 (c) $35 + j14$

Problem6:

Problem7: $v_s(t) = 1.414 \sin(200t - 45^\circ)\text{ V}$

Problem8: $Z_T = 6.83 + j1.094\ \Omega = 6.917 \angle 9.1^\circ\ \Omega$ $I = 17.35 \angle 0.9^\circ\text{ A}$

Problem9: Proof

Problem10: $C = 235 \text{ pF}$

HW10:

Problem1: $i(t) = 1.9704\cos(10t+5.65^\circ) \text{ A}$

Problem2: $V = 124.08\angle -154^\circ \text{ V}$

Problem3: $i_o(t) = 2.828 \cos(2t + 45^\circ) \text{ A}$

Problem4: $i_1(t) = 2.741\cos(4t-41.07^\circ) \text{ A}$, $i_2(t) = 4.114\cos(4t+92^\circ) \text{ A}$.

Problem5: $i_o(t) = 8 + 1.5811 \cos(4t - 71.57^\circ) \text{ A}$

Problem6: $i(t) = [989.1\cos(10t+21.47^\circ) + 499\sin(4t+176.57^\circ)] \text{ mA}$

Problem7: $i(t) = 8.944 \sin(200t + 56.56^\circ) \text{ A}$

Problem8: a) $Z_n = Z_{th} = 22.36\angle -63.43^\circ \Omega$ $V_{th} = -50\angle 30^\circ \text{ V}$ $I_n = 2.236\angle 273.4^\circ \text{ A}$

b) $Z_n = Z_{th} = 10\angle 26^\circ \Omega$ $V_{th} = 33.92\angle 58^\circ \text{ V}$ $I_n = 3.392\angle 32^\circ \text{ A}$

Problem9: $i_o(t) = 89.45 \cos(10^4 t + 3.44^\circ) \mu\text{A}$

Problem10: $v_o(t) = 3.578 \cos(1000t + 26.56^\circ) \text{ V}$